Answer: C

15 $$\frac{GM_S m}{r^2} = mr\omega^2$$

$$\frac{GM_S}{r^3} = \frac{4\pi^2}{T^2}$$

$$\therefore r^3 \propto T^2$$

$$\left(\frac{1.43\times10^{12}}{7.78\times10^{11}}\right)^3 = \left(\frac{T_{Saturn}}{11.9}\right)^2$$

$$T_{Saturn} = 29.7 \text{ years}$$

Answer: C

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